

Foundations of Artificial Intelligence

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Exercise Sheet 5

Due: March 29, 2026

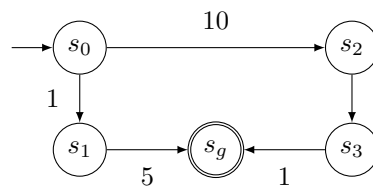
Please note the submission rules at the end of this exercise sheet. Non-adherence to these rules may lead to your submission not being marked.

Exercise 5.1 (1+1 marks)

Execute

- (a) greedy best-first search ($f(n) = h(n.state)$) and
- (b) A* ($f(n) = g(n) + h(n.state)$)

without reopening in the state space depicted below. As heuristic, use the perfect heuristic h^* .



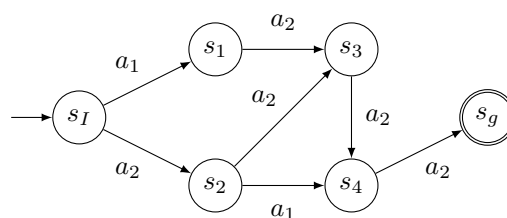
Describe the execution of both search algorithms with the following schema:

1. expanding s_{10} : $open = \langle s_{11}(f = 5), s_{12}(f = 6) \rangle$, $closed = \{s_{10}(g = 0)\}$
2. expanding s_{11} : $open = \langle s_{13}(f = 4), s_{12}(f = 6), s_{16}(f = 10) \rangle$, $closed = \{s_{10}(g = 0), s_{11}(g = 3)\}$
- ...
- i . expanding s_i : found goal with cost x

Note: This schema uses fictitious states and numbers that don't correspond to the given state space.

Exercise 5.2 (2 marks)

Consider the following state space $\mathcal{S}$ where $cost(a_1) = cost(a_2) = 2$:



Define an admissible heuristic h for $\mathcal{S}$ such that IDA* behaves in the following way:

- IDA* performs at least 4 iterations and
- exactly one new path is reachable within the new f limit in each iteration (simulating worst-case behavior of IDA*).

Provide your heuristic h and for each iteration the f limit as well as the newly explored path.
Hint: There are solutions which use only natural numbers (including 0) as heuristic values. To satisfy the second requirement, expanded nodes must have distinct f -values.

Exercise 5.3 (2 marks)

In Chapter B14 we showed that A^* with reopening is optimal when using admissible heuristics. If you would try to apply the proof to weighted A^* instead of A^* , where would it fail? Justify why it fails at the point you specify and briefly explain why it does not fail earlier.

Exercise 5.4 (2+2 marks)

Show with a counterexample that the following two statements are not correct:

- (a) A^* without reopening using an admissible but inconsistent heuristic is optimal.
- (b) A^* without reopening using a consistent but inadmissible heuristic is optimal.

For each subtask, the counterexample must consist of a state space and a heuristic. Provide the state space as a directed graph, similar to the “bounded inc-and-square” state space introduced in Chapter B1. Add the action costs and the heuristic values as additional labels to the edges and nodes, respectively, you don’t need to write them down separately. Show that the heuristic has the required properties, and show that A^* is not optimal. The solution found should be suboptimal independently of tie-breaking, that is, it should not depend on which states with the same f -value are expanded first.

Hint: For one of the tasks (a) or (b), the counterexample requires multiple paths to the same state. For the other, the counterexample requires more than one goal state.

Submission rules:

- Submit one solution in groups of up to two students, stating the names of all group members in the submission. All group members are equally responsible for all submitted exercises.
- Create a single A4 PDF file for all non-programming exercises. Solutions in $\text{\LaTeX}$ are preferred, but clearly legible scans in A4 with reasonable contrast and file size are accepted as well.
- In case there are programming exercises, upload a ZIP file (ending .zip, .tar.gz or .tgz; *not* .rar or anything else) including your PDF and those code textfiles required by the exercise. Put your names in a comment on top of each code file. Make sure your code compiles and test it. Code that does not compile or which we cannot successfully execute will not be graded.
- Include a statement on scientific integrity at the end of your submission (see below).

Statement on scientific integrity:

- All submitted solutions must be the product of your own work.
- If your written solutions were produced using any help (e.g., from peers or a large language model), list in what way these people or tools contributed to the solutions.
- We do not mark submissions that lack a statement on scientific integrity.
- Examples:
 - “We solved all exercises on our own.”
 - “We used ChatGPT to explain exercise 7 which we did not understand.”
 - “We had no time to solve this exercise sheet and copied all solutions from our friends.”
(If this is the case, thanks for your honesty but this is plagiarism, results in a warning and no marks. It is preferable to submit no solution at all.)